Human vs. AI in Symbolic Music:
An Online Listening Study on Bach-Style Chorales

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Introduction & Method

We conducted a music perception study to evaluate human vs. AI discrimination in Bach-style chorales. Participants ($N = 295$) listened to **20 isolated musical phrases** and categorized each as **AI**- or **human-composed**. AI snippets were generated using *NotaGen* [1], a state-of-the-art symbolic music generation model. All musical snippets (both human and AI) were manually curated. Participation was open to the public with **no prior music qualification required**. Prior to and following the experiment, we collected metadata such as age, musical background, and self-reported confidence levels in order to better contextualize participant performance.

Experiment Design

Pre-Survey → **20 Listening Trials** → **Post-Survey**

(Order randomized)

Stimuli

- **Balanced Set:** 10 Bach chorale phrases vs. 10 *NotaGen* phrases
- **Audio Uniformity:** All excerpts rendered with an identical MIDI harpsichord soundfont
- **Selection:** All snippets were manually curated

Measured Parameters

- **Demographics:** Age, weekly AI usage (Yes/No).
- **Musical Background:** Plays instrument (Yes/No), sings in choir (Yes/No), self-reported experience (4-point Likert scale).
- **Confidence:** Pre- and post-experiment self-ratings (4-point Likert scale).

Overview and Results

Participant Overview

- **Sample Size:** $N = 295$ participants (primarily German, aged 15–86).
- **Musical Background:** $\approx$ 90% play an instrument; $>$ 50% sing in a choir.
- **AI Usage:** $\approx$ 50% use AI multiple times per week.

Participant Feedback[1]

- *"frighteningly difficult 🙂"*
- *"It was enormously difficult to hear a difference. I was constantly guessing. If I guessed something correctly, it merely was a gut instinct. [...]"*
- *"It was a very interesting experience because I realized, how damn hard it is to me distinguish AI compositions from human compositions."*
- *"The snippets are relatively short. Longer snippets could create a higher hit rate, I guess..."*
- *"It's really exciting because it's really difficult to decide whether it was AI or human after hearing it only once. Sometimes I would have needed it a second time."*

[1]Quotes were translated from German to English

Age Distribution

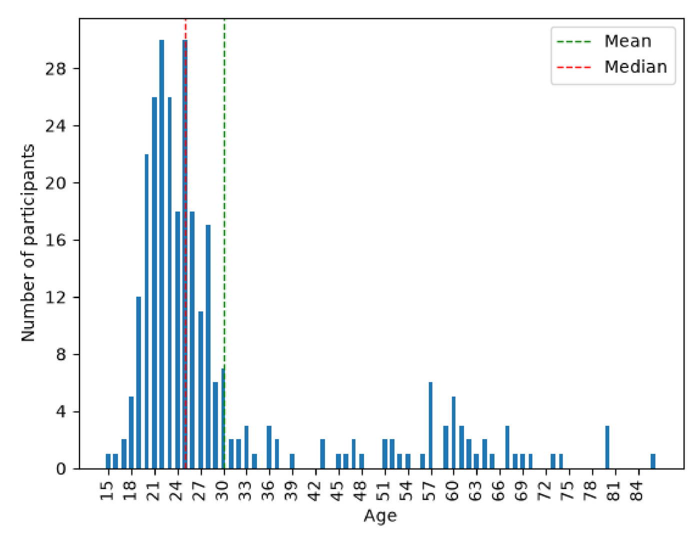

Figure 1: *Participant age distribution ($N = 295$)*

Performance by Choir Experience

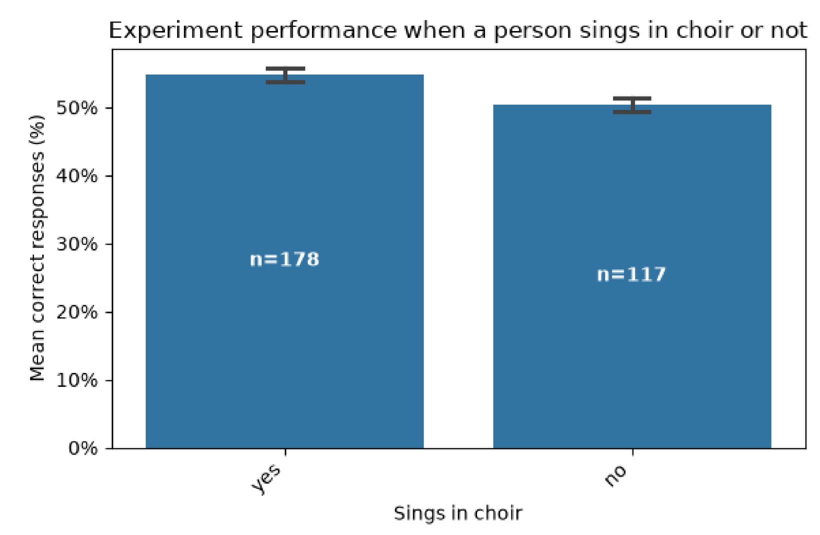

Figure 3: *Accuracy comparison by participants' choir experience*

Performance by Musical Experience

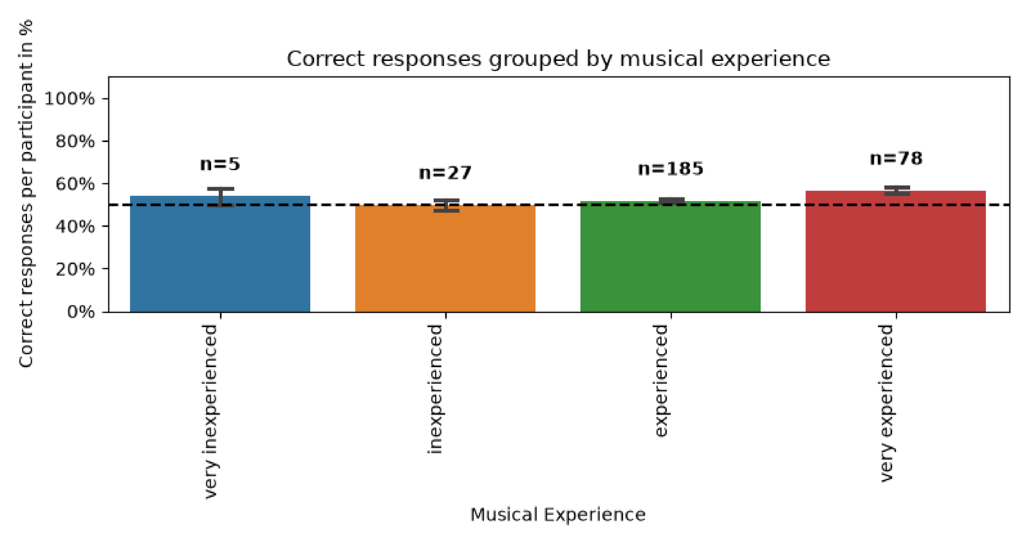

Figure 2: *Discrimination accuracy across participants' prior musical experience*

Confidence Shifts

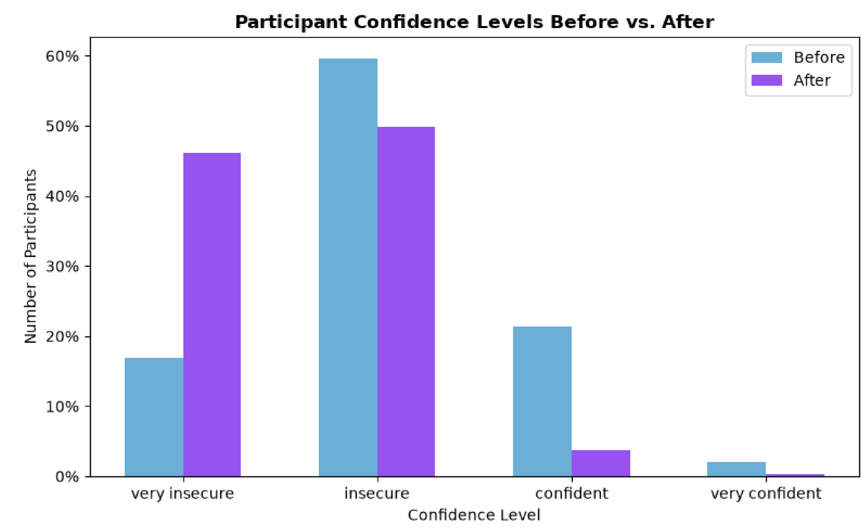

Figure 4: *Self-reported confidence pre- vs. post-trial*

Overview: Performance Per Participant

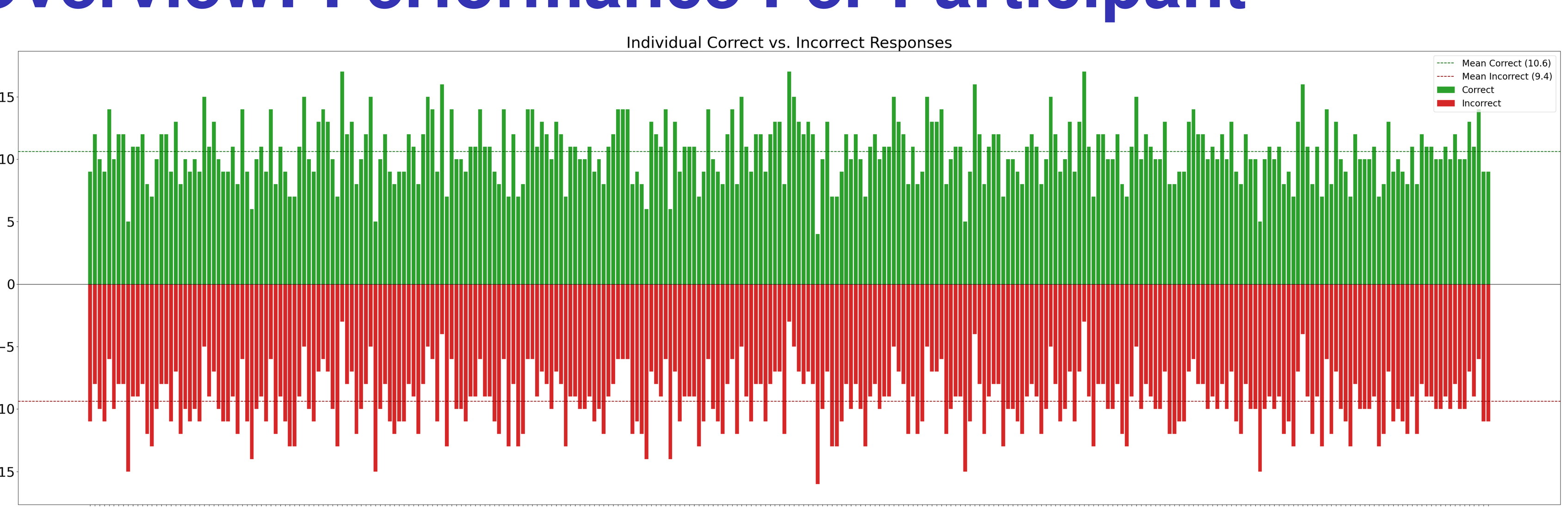

Figure 5: *Individual participant accuracy scores across all trials*

Stimuli Discrimination Accuracy

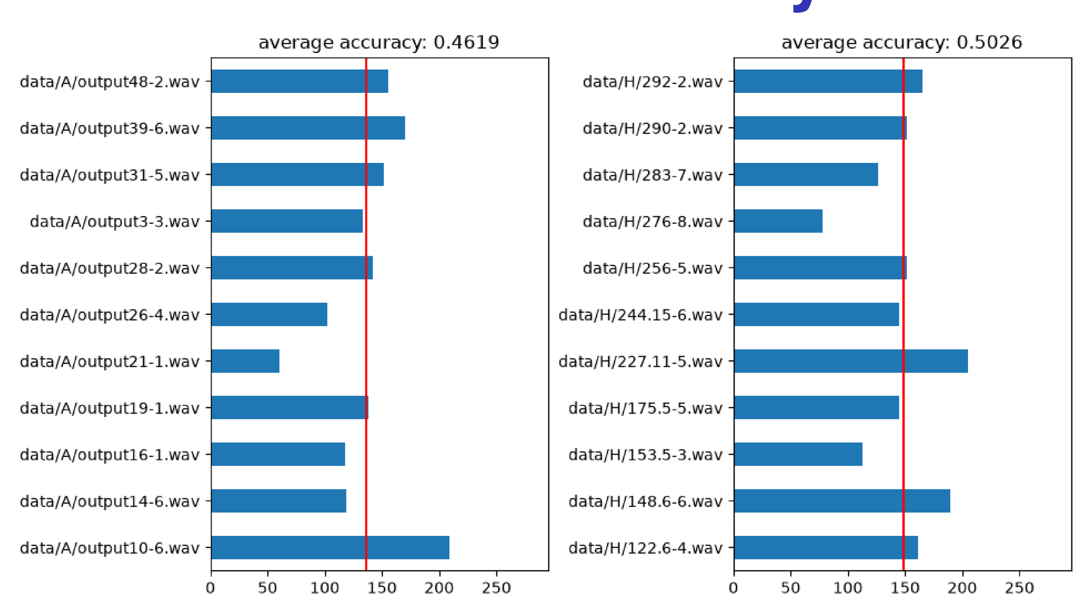

Figure 6: *Overview of individual stimulus discrimination. Left plot shows AI snippets. Right plot shows human-composed snippets.*

References

[1] Yashan Wang et al. *NotaGen: Advancing Musicality in Symbolic Music Generation with Large Language Model Training Paradigms.* Mar. 2025. DOI: 10.48550/arXiv.2502.18008. arXiv: 2502.18008 [cs]. URL: http://arxiv.org/abs/2502.18008 (visited on 01/21/2026).

Try the Experiment

Can you distinguish AI from human-composed?

← Scan to test your skills in our study

jasonvla.github.io/ai-chorales_website

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